

User Defined Functions/Methods : Questions and programs

1. What is function header and body ?
2. Do non static members of a class belong to class or object of that class ?
3. what is return type ? what does it mean when you write 'void' as return type of a function ?
4. what are formal and actual parameters ?
5. what is call by value and call by reference ?
6. give one example of parameterized and non parameterized function.
7. What is method signature
8. what are pure and impure functions ?
9. what is state and behaviour of a class
10. what is recursion ?
11. what is nested call ?
12. how many values can a function return ?
13. how many return statements can be there in a function
14. Can we return a value of type int from a function whose return type is float
15. can we call a non static function inside and a non static function without another object
16. is it possible to pass an object as parameter ?
17. is function and method same thing ?
18. what are instance, class and local variables ?
19. how do we implement polymorphism in a class ?
20. write prototype of a function that takes integer and returns a character

Programming Questions: (Questions that do not mention variable names, class names- make assumptions)

Every program must contain the main function to call other functions even if it's not mentioned in question

- 1. Design a function** `int Max()` to accept **two** integer arguments and return the largest integer.
- 2. Design a function** `int Min()` to accept **three** integer arguments and return the smallest.
- 3. Create a function** `void process(char ch, int x, int y)` to accept an arithmetic operator (+, -, /, *) in `ch` and two integers `x` and `y`. The function should perform the corresponding arithmetic operation and print the final result.

Also, write a main function to input an arithmetic operator (`ch`) and two integers (`x` and `y`).

Based on the operator stored in `ch`, perform the operation by invoking the function `process()` and print the output.

4. Define a method void printDetails(String pname, float pr, int qty) to read name of the product (pname), unit price of the product (pr) and quantity of the product (qty). calculate total cost of the product, discount of 20% if total cost in more than Rs. 10000/- , otherwise no discount. also write the main function to input name, quantity, and price, the call the function printDetails()

5. Design a function maxChar(char, char) – return the smallest character from the arguments.

6. Create a function int counter(long num, int d) to return occurrence of d in num. (hint: digit extraction)

7. Define a function maxDigit(int num) to find and return the greatest digit from the argument

8. Write a function comparison that takes two char parameters and returns 0 if both parameters are equal otherwise it should return -1

9. Design a function int factorial(int num) to calculate and return factors of num

10. Design a function int power(int b, int e) to return b^e without using Math.pow()

11. Write a function int check(char ch) to return 1 if ch is in uppercase otherwise return 0

12. Write a function char convert() that takes a character from user and returns its opposite case character

13. Design a function int Prime(int n) to return 1 if n is prime otherwise return 0. call the function and check if returned value is 1 or 0, based on that print the message.

14. Write a program to print all prime numbers between two user given numbers using these functions

void showPrime(int a, int b)

boolean isPrime(int n)

write the main function and call the functions

15. Write a function int Perfect(int Y) to return 1 if Y is perfect otherwise returns 0.

Perfect number : a number whose sum of factors (except itself) is equal to the number

16. write a menu driven program to call overloaded function area() to print area of circle, square or rectangle

area(double r) – receives radius and prints area of the circle

area(int length, int breadth) to print area of rectangle

area(int side) to print area of square

17. define a function int isPrime(int n) to return 1 if n is prime otherwise return 0

Now, generate fibonacci numbers between 1 and 100 which are prime using the function isPrime(int n)

18. Design a function long product(long num) to return product of first and last digit of num

19. Using a suitable function of your choice print the sum of the following series

$$S = \frac{3!}{1} + \frac{4!}{2} + \frac{5!}{3} + \frac{6!}{4} + \frac{7!}{5} + \dots + \frac{n+2!}{n}$$

20. Define a class Equation to print the result of the following equation:

$$\frac{n!}{r!(n-r)!}$$

member functions:

int fact(int n) to return factorial of n

void printResults(int r, int n) to print the result of the above function using int fact()

Write the main function to call

21. create the following functions :

int fact(int y) to return factorial of y

void printSpecial(int s, int l) by invoking fact() print all special numbers between s and l

write the main function to execute

22. Design a function printSalary(String name, float rate, float hour, int days)- to read name of the worker(name) , payment rate per hour (rate), total number of hours worked, and number of days worked (days)

calculate salary using this formula : rate x hours x days

23. Write a function void sum(int X) to perform following tasks:

i. Reverse the X and store in another variable Y

ii. Add X and Y and store result in Z

iii. Find sum of all digits of Z

iv. Print X, Y, Z and S with suitable messages

24. Design a function long Merge(long x, long y) that accepts numbers in x and y. Merge the numbers stored in x and y (numerically) and return merged number

25. Design a function void findVowel(char ch) that takes a character in ch and find its nearest vowel if ch is not a vowel, otherwise print itself

26. Create a function void Number(long num) to print the digits of the number using words

27. Write a class to design the following functions:

void change(char, char) : to change case of each character

void convert(int, int) to print character equivalents of each integer

28. Design a class to overload the given function as follows

double volume(double) find volume of a cube using side³

double volume(double, double, double) – to find and return volume of cuboid using length* breadth* height

double volume(float) – find and return volume of a cylinder using $(4/3) \pi r^3$

29. Write a program to check whether n and m are twin primes or not using these functions:

boolean prime(long num) : to return true if num is prime otherwise false

void twinPrime(int m, int n): to check if m and n both are prime or not using prime() , if yes, print
“Twin Prime”

30. Design a class to overload a function area() as follows:

double area(double a, double b, double c) to return area of the scalene triangle using the formula

$area = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$

double area(int a, int b, int height) to find area of a trapezium using formula

$area = (1/2)height (a+b)$

double area (double diagonal1, double diagonal2) with two double arguments, returns the area of a rhombus using the formula

$area = (1/2) (diagonal1 \times diagonal2)$

31. Design a class to overload a function SumSeries() as follows

void SumSeries(int n, double x) – with one integer argument and one double argument to find and display the sum of the given series

$$S = \frac{x}{1} - \frac{x}{1} + \frac{x}{2} - \frac{x}{3} + \frac{x}{4} - \frac{x}{5} + \dots \cdot \frac{x}{n}$$

void SumSeries()- find and display the sum of the following series:

$S = 1 + (1 \times 2) + (1 \times 2 \times 3) + (1 \times 2 \times 3 \times 4) + \dots (1 \times 2 \times 3 \times 4 \times 5 \dots \times 20)$

32. define a class ass follows:

class name: Interest

Data members: int p

int r

int t

int si

member methods: accept() to take principle, rate and time from user in p, r and t

calculate() to find simple interest using $p \times r \times t / 100$

display() to show the output

Call the functions in main functions

33.

Define a class to overload the method display() as follows:

void display(): To print the following format using nested loop.

1 2 1 2 1

1 2 1 2 1

1 2 1 2 1

void display (int n, int m) : To print the quotient of the division of m and n if m is greater than n otherwise print the sum of twice n and thrice m.

double display (double a, double b, double c) – to print the value of z where

$$z = p \times q$$

$$p = \frac{a+b}{c} \quad q = a + b + c$$

34.

Define a class to overload the method perform as follows:

double perform (double r, double h) to calculate and return the value of
Curved surface area of cone

$$CSA = \pi r l \quad l = \sqrt{r^2 + h^2}$$

void perform (int r, int c) Use NESTED FOR LOOP to generate the
following format

r = 4, c = 5 output – 1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

void perform (int m, int n, char ch) to print the quotient of the division of m and
n if ch is Q else print the remainder of the
division of m and n if ch is R